

| Question |     |       |    | Marking details                                                                                                                                                                                                                                                                                                             | Marks available |          |          |           | Maths    | Prac     |
|----------|-----|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |    |                                                                                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total     |          |          |
| 6        | (a) | (i)   |    | $T$ tension in the string <b>and</b> $mg$ weight of mass (gravitational force or gravity)                                                                                                                                                                                                                                   | 1               |          |          | 1         |          |          |
|          |     | (ii)  |    | $T$ does not have a component tangential to the arc (1)<br>Component of $mg$ tangential to the arc is $mg \sin \theta$ , (1) this is in the opposite direction to $s$ (or $\theta$ ) <b>and</b> so the negative sign (1)                                                                                                    |                 | 3        |          | 3         | 1        |          |
|          |     | (iii) |    | acceleration = $\frac{-mg \sin \theta}{m} = -g \sin \theta$<br>= $-g \theta$ (1)(using the approximation)<br>$\theta = \frac{s}{l}$ or $\sin \theta = \frac{s}{l}$ (1) = $-\frac{gs}{l}$                                                                                                                                    |                 | 2        |          | 2         | 2        |          |
|          |     | (iv)  |    | Acceleration $\propto \theta$ (or $s$ ) measured [from a fixed position] (1)<br>and opposite in direction (-ve) so SHM (1)                                                                                                                                                                                                  |                 |          | 2        | 2         |          |          |
|          | (b) | (i)   | I  | Substitution: $T = 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{1.2}{9.81}}$ (1)<br>= 2.20 [s] (1)                                                                                                                                                                                                                            | 1               | 1        |          | 2         | 2        |          |
|          |     |       | II | $f = \frac{1}{T} = 0.45[5]$ [Hz] <b>ecf</b> (1)                                                                                                                                                                                                                                                                             |                 | 1        |          | 1         | 1        |          |
|          |     | (ii)  |    | For maximum distance along the arc $\theta_{\max} = 0.067$ ,<br>also $\sin \theta_{\max} = 0.067$<br>As $\theta_{\max} = \sin \theta_{\max}$ (i.e. for the largest value of $\theta$ ) (1)<br>then $\sin \theta$ is equal to $\theta$ for all $\theta$ , and approximation holds for SHM.<br>System oscillates with SHM (1) |                 |          | 2        | 2         | 1        |          |
|          |     |       |    | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                     | <b>2</b>        | <b>7</b> | <b>4</b> | <b>13</b> | <b>7</b> | <b>0</b> |